COLLEGE COMPUTER LABORATORY NETWORK (DHCP + DNS + FTP)

A college wants to establish a computer laboratory with 20 student computers, 2 faculty computers, and a central server. Currently, the network administrator manually configures IP addresses on every computer. Students also face difficulties accessing academic resources because they must remember server IP addresses, and faculty members frequently use USB drives to distribute laboratory manuals and assignments.

As a network administrator, you are assigned to design and implement an integrated network using **Cisco Packet Tracer** that simplifies IP configuration, enables domain name resolution, and facilitates file sharing.

Question:

Design and configure a network using Cisco Packet Tracer by integrating **DHCP (Dynamic Host Configuration Protocol), DNS (Domain Name System), and FTP (File Transfer Protocol)** on a central server. Configure DHCP to assign IP addresses automatically to all client computers, DNS to resolve a user-friendly domain name to the server's IP address, and FTP to allow authorized students and faculty members to upload or download academic resources according to their access permissions. Demonstrate the successful operation of all three services through appropriate connectivity tests, domain name resolution, and FTP file-transfer operations.

Requirements:

1. Design the network topology using a switch, a central server, and client computers.
2. Configure the IP addressing scheme and DHCP service.
3. Configure a DNS record for the college server.
4. Configure FTP accounts with appropriate access permissions.
5. Verify connectivity using ping.
6. Test DNS resolution and FTP file transfers.
7. Document the configuration and testing results.

Brief Overview

A college computer laboratory network connects student computers, faculty computers, and a central server through a single switch. The central server provides three services: DHCP to assign IP addresses automatically, DNS to resolve user-friendly domain names into the server’s IP address, and FTP to share academic resources such as laboratory manuals and assignments.

DHCP eliminates the need to configure each computer manually. It follows the DORA process (Discover, Offer, Request, Acknowledge) and also provides the subnet mask, default gateway, and DNS server address.

DNS allows users to access the server using a domain name such as lab.college.edu instead of remembering its IP address.

FTP uses TCP port 21 for control and, in active mode, TCP port 20 for data transfer. It allows authorized users to upload and download files based on the permissions assigned to their accounts.

Together, these services eliminate manual IP configuration, simplify server access through domain names, and reduce the need for USB drives to share academic resources.

Advantages of the Integrated Network

▪ Automatically assigns IP addresses to all computers, which removes manual configuration and prevents duplicate addresses.
▪ Allows users to reach the server by a domain name instead of remembering its IP address.
▪ Makes file sharing easy, since laboratory manuals and assignments are stored centrally and no USB drives are needed.
▪ Provides access control, because each FTP account has its own permissions (for example, faculty can upload and manage files while students can only download).
▪ Makes network management easier, because addressing, naming, and file storage are all controlled from one central server.
▪ Easy to expand, because new computers receive their configuration automatically when connected to the switch.

Disadvantages of the Integrated Network

▪ The central server is a single point of failure, so if it fails, DHCP, DNS, and FTP all stop working together.
▪ Standard FTP sends usernames, passwords, and data in plain text, so it is not secure without protection such as FTPS or SFTP.
▪ The server must keep a static IP address and be carefully configured so that it does not conflict with the DHCP pool.
▪ Network performance depends on the capacity of the server and the switch, and an overloaded server can slow every service.
▪ Packet Tracer sets FTP permissions per user, not per folder, so access control is less detailed than on a real server.
▪ The switch has a limited number of ports, so a larger lab would need additional switches.

Simulation / Design in Cisco Packet Tracer

This section presents the college computer laboratory network designed and simulated in Cisco Packet Tracer. The network consists of 20 student computers (STU1 to STU20), 2 faculty computers (FAC1 and FAC2), and one central server connected to a 2960-24TT switch. The server acts as the DHCP server, the DNS server, and the FTP server. The computers obtain their IP configuration automatically, reach the server using the domain name lab.college.edu, and upload or download files according to their FTP account permissions

Network Diagram

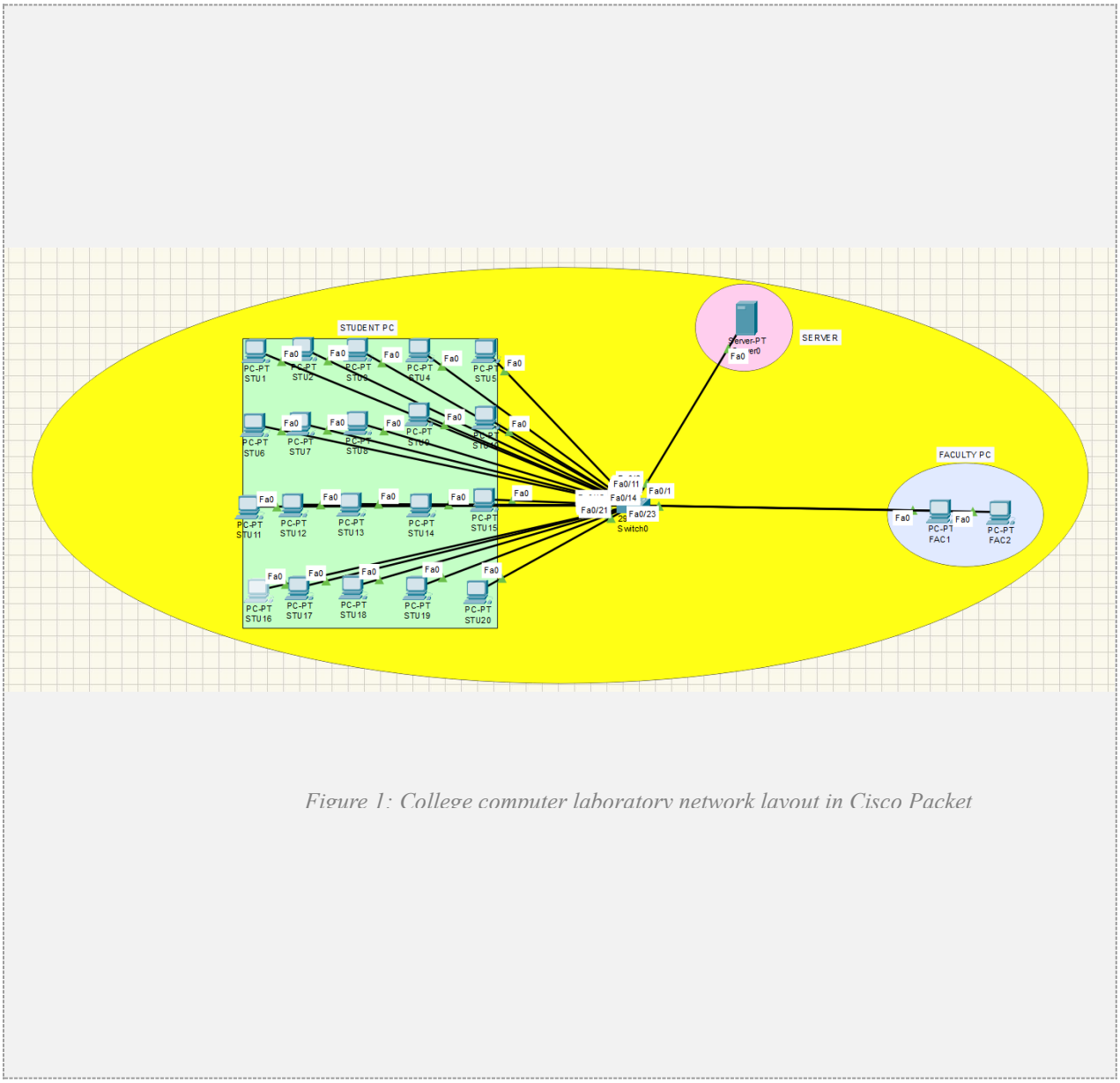


Figure 1: College computer laboratory network layout in Cisco Packet

Device Configuration

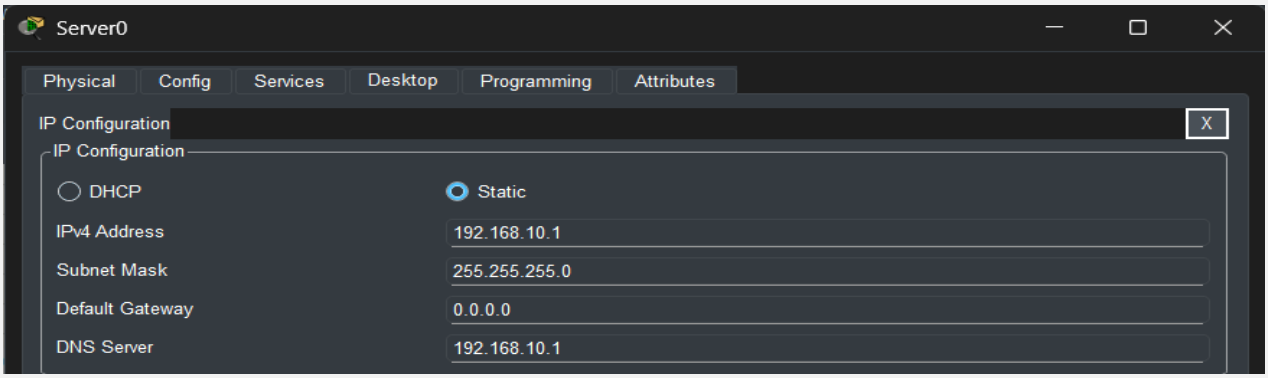


Figure 2.1: Static IP configuration of the server (192.168.10.1)

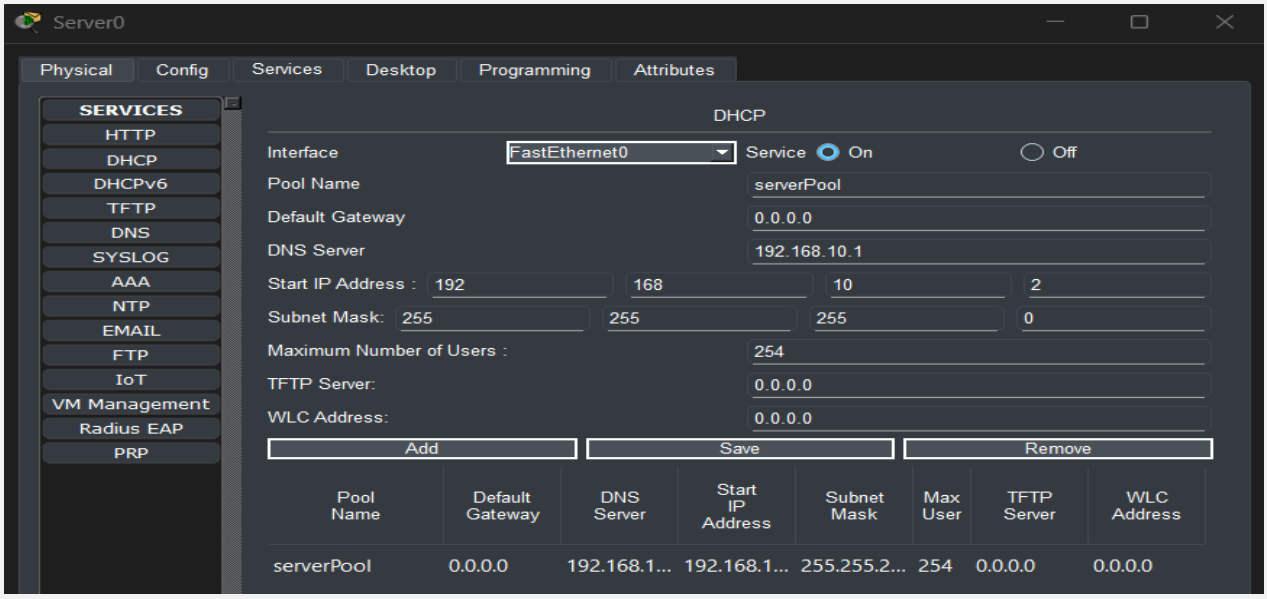


Figure 2.2: DHCP service configuration window of the server (serverPool)

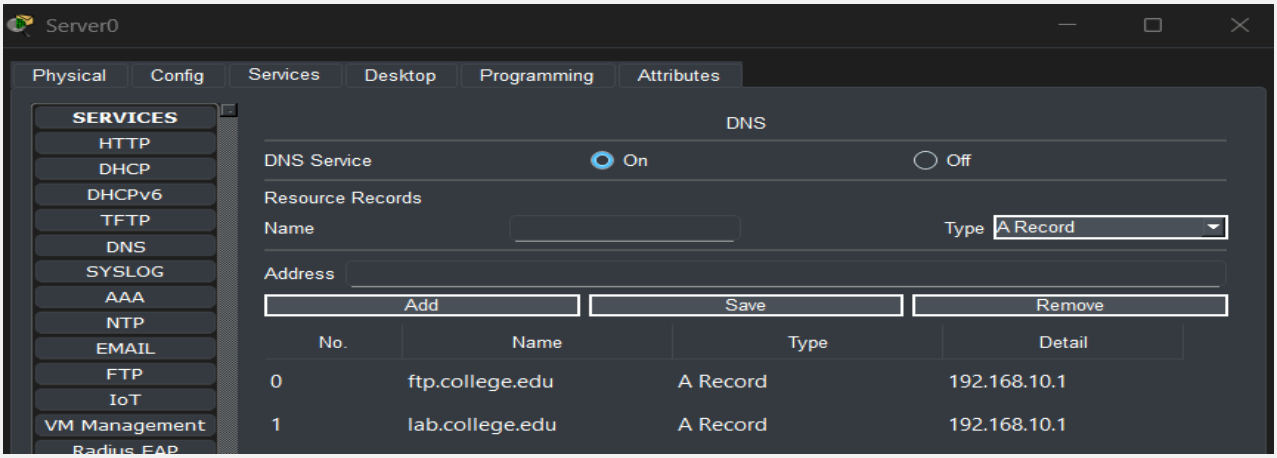


Figure 2.3: DNS service configuration with the domain name record

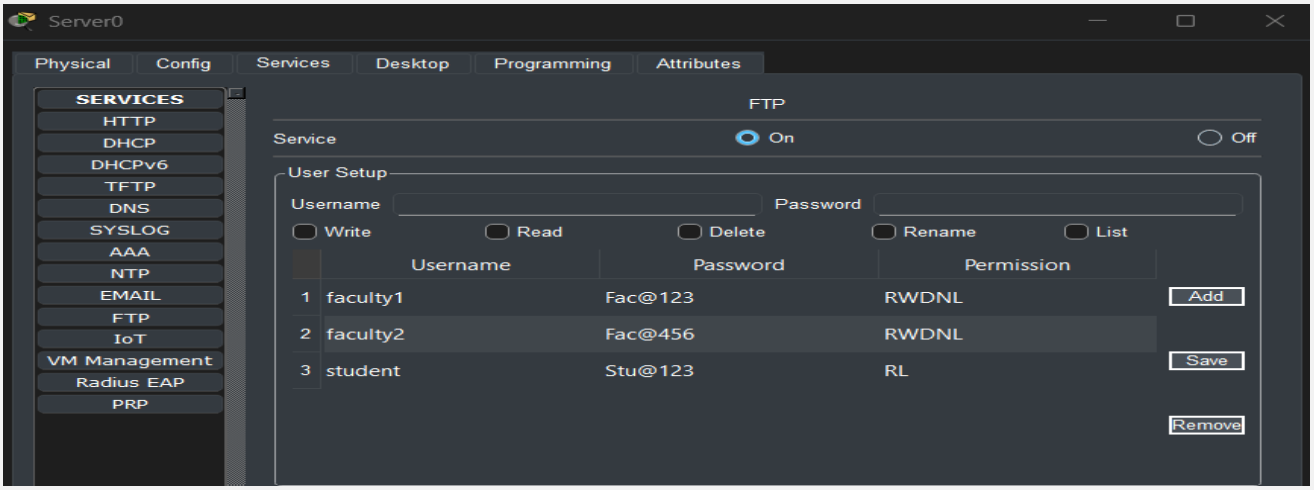


Figure 2.4: FTP service configuration with the user accounts and permissions

Connectivity Test (Ping / DNS / FTP / Simulation Result)

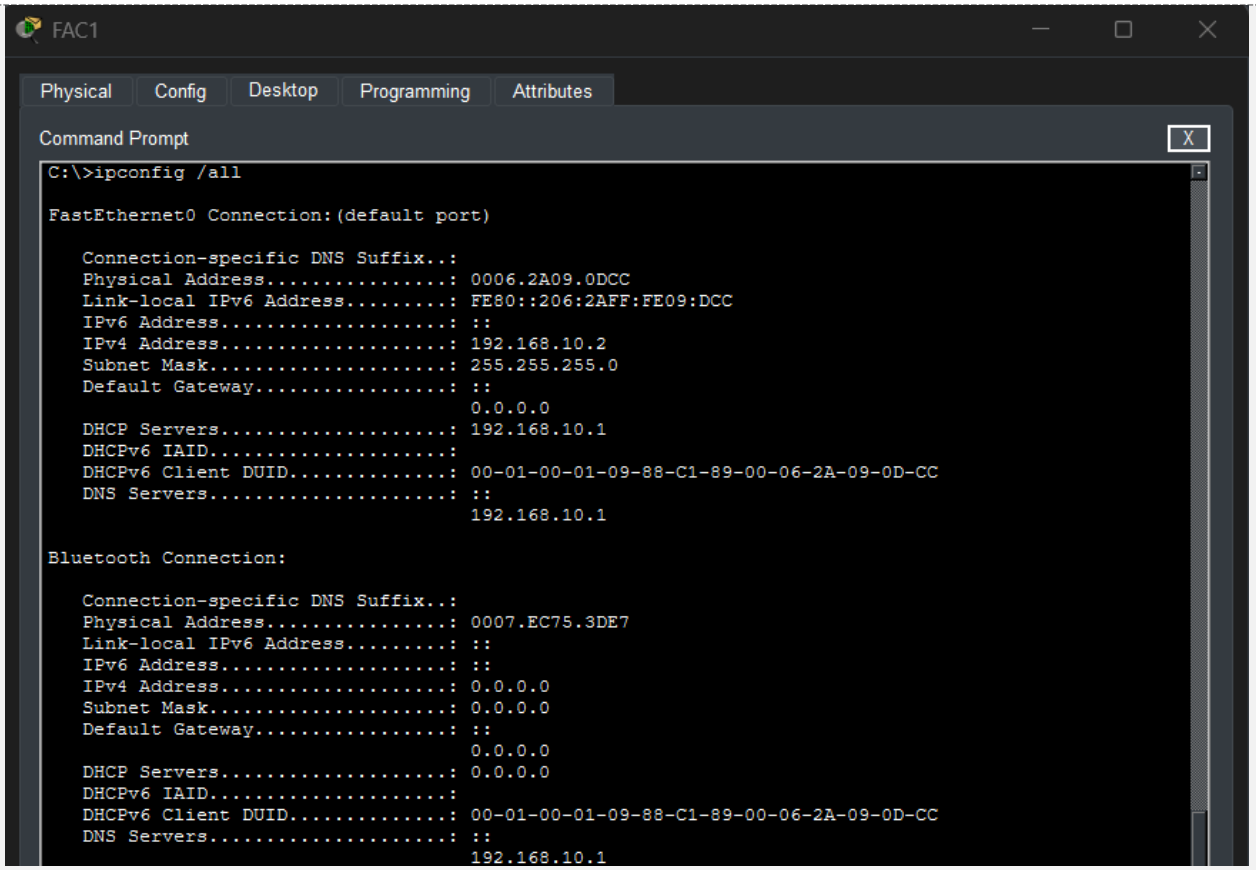


Figure 3.1: ipconfig /all output showing the DHCP-assigned configuration

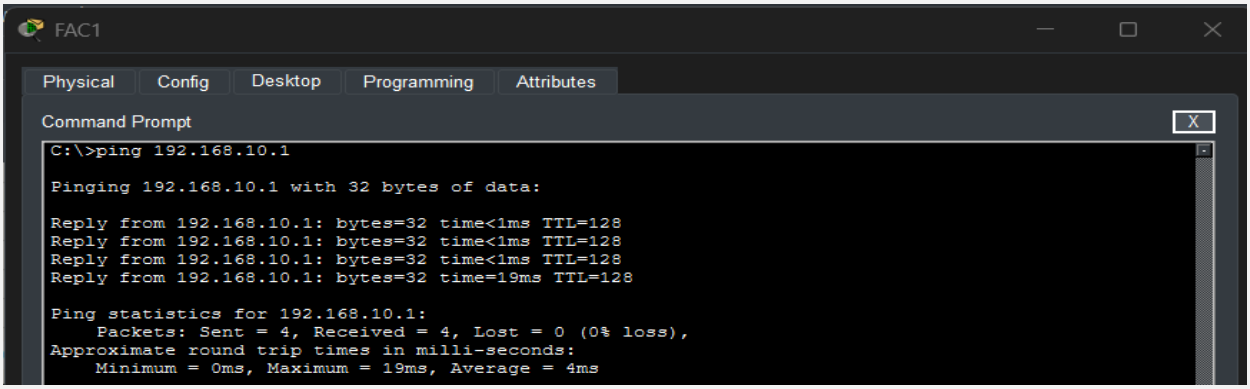


Figure 3.2: Successful ping from a PC to the server

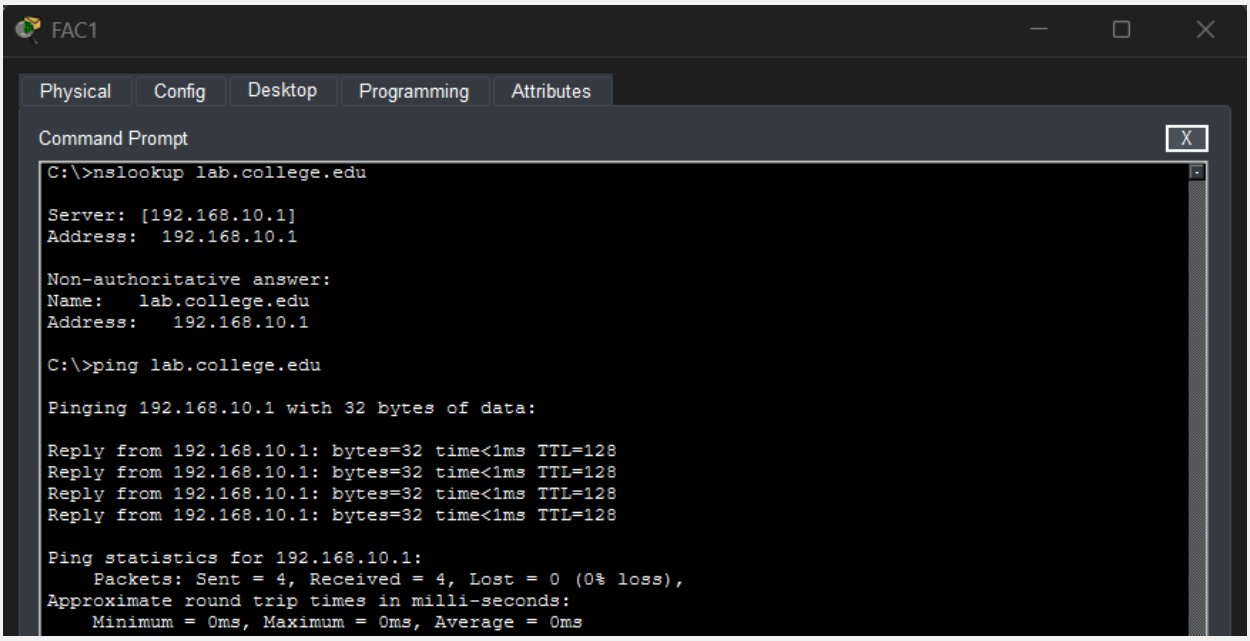


Figure 3.4: Successful nslookup and ping using the domain name

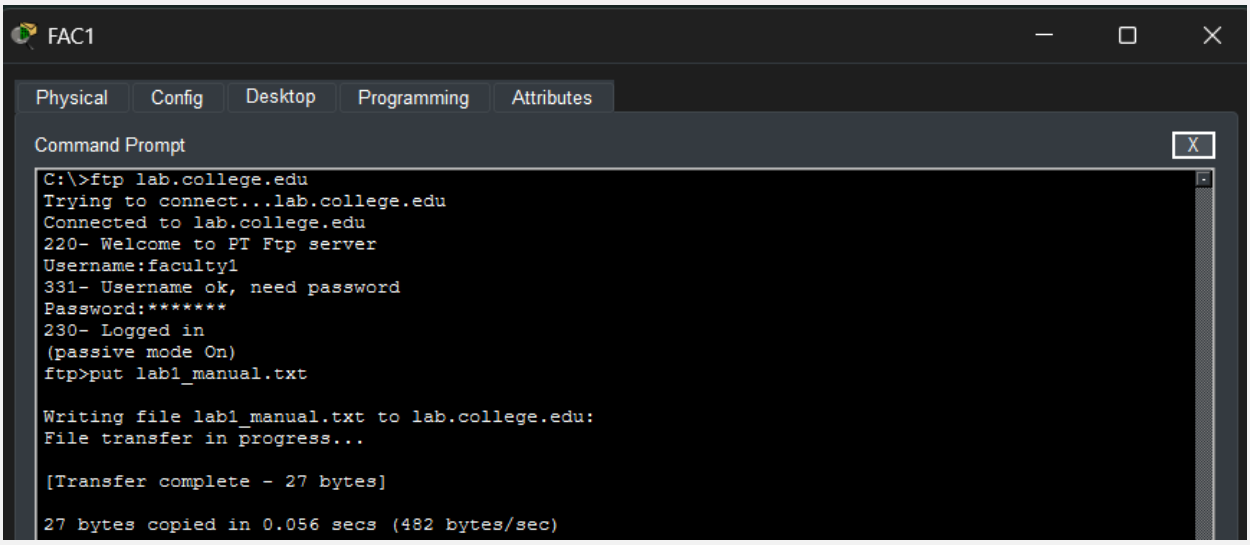


Figure 3.5: Successful FTP login and file upload by the faculty user

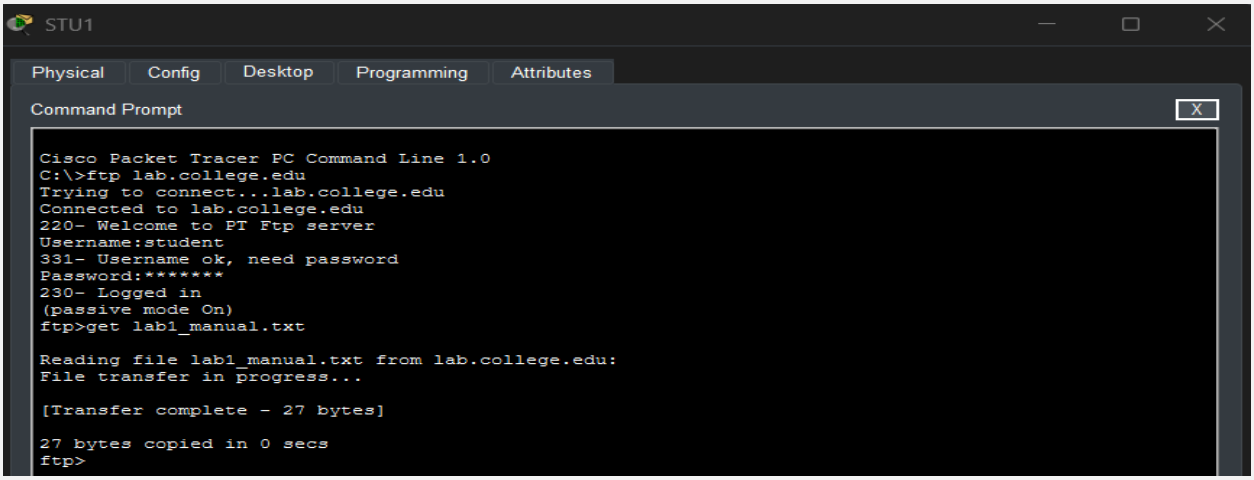


Figure 3.6: Successful FTP login and file upload by the student

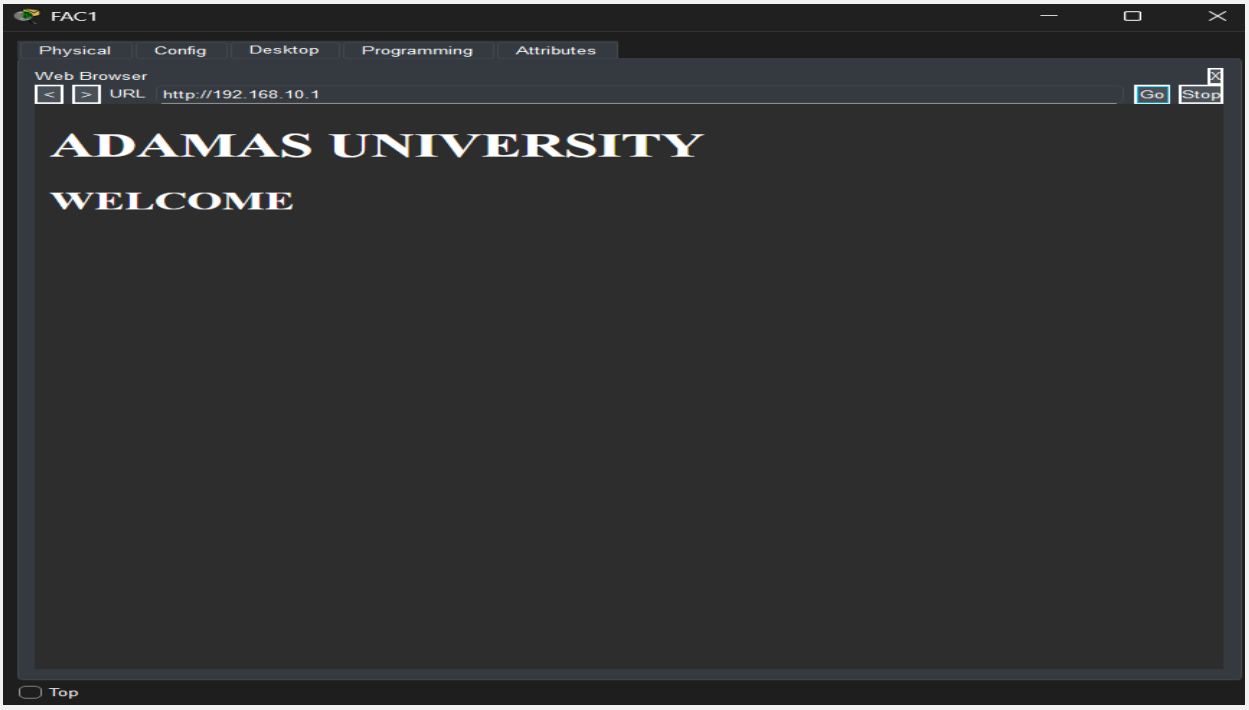


Figure 3.7: Successful access to the college server web page (192.168.10.1)

Process of Designing the Network in Packet Tracer

▪ Open Cisco Packet Tracer and create a new blank project to begin designing the college laboratory network.

▪ Select a 2960-24TT switch from the Network Devices category and place it at the center of the workspace. Its 24 ports are enough for 20 student PCs, 2 faculty PCs, and 1 server.

▪ From the End Devices category, place one Server-PT and 22 PC-PTs. Name the student PCs STU1 to STU20 and the faculty PCs FAC1 and FAC2. A few PCs can be placed first and then copied and pasted (Ctrl+C and Ctrl+V) to save time.

▪ Select Copper Straight-Through cable from the Connections category and connect the server to port Fa0/1 of the switch. Then connect each PC, through its FastEthernet0 port, to the remaining ports (Fa0/2 to Fa0/23). Wait until the link lights turn green, which shows that the physical connections are active.

▪ The addressing plan uses the network 192.168.10.0/24, as shown below.

Item	Value
Server (static)	192.168.10.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.10.1
DHCP Pool Start	192.168.10.2
Maximum Number of Users	254
Domain Name	lab.college.edu → 192.168.10.1

▪ Give the server a static IP address, because the PCs depend on it for DHCP, DNS, and FTP. Open Server → Desktop → IP Configuration, select Static, and enter the following values.

Field	Value
IPv4 Address	192.168.10.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.10.1

▪ The default gateway is left as 0.0.0.0 because all devices are in the same network and no router is used. The DNS server field points to the server itself, since it acts as the DNS server.

▪ Configure the DHCP service by opening Server → Services → DHCP and setting the service to On. Keep the default pool name (serverPool), enter the following details, and click Save.

Field	Value
Pool Name	serverPool
Default Gateway	0.0.0.0
DNS Server	192.168.10.1
Start IP Address	192.168.10.2
Subnet Mask	255.255.255.0

Field	Value
Maximum Number of Users	254

▪ The pool starts from 192.168.10.2, so the server's own address (192.168.10.1) is never handed out to a client. The DNS server address in the pool ensures that every PC automatically receives 192.168.10.1 as its DNS server.

▪ Configure the DNS service by opening Server → Services → DNS and turning the DNS service On.

▪ Add a record for the server. Enter the name lab.college.edu, select the type A Record, and enter the address 192.168.10.1. Click Add to save the record. Optionally, add a second record, ftp.college.edu, pointing to 192.168.10.1.

▪ Configure the FTP service by opening Server → Services → FTP and setting the service to On.

▪ Delete the default user cisco, because it has full permissions and is a security risk.

▪ Create the user accounts by entering a username and password, ticking the required permissions, and clicking the + button for each account.

Username	Password	Permissions	Role
faculty1	Fac@123	Write, Read, Delete, Rename, List	Uploads manuals and assignments and can manage files
faculty2	Fac@456	Write, Read, Delete, Rename, List	Same as faculty1
student1	Stu@123	Read, List	Can only download files

▪ More student accounts can be added, or one student account can be shared. Packet Tracer sets FTP permissions per user, not per folder, which is enough to show the idea of access control.

▪ Make the computers request an address. On every PC, open Desktop → IP Configuration and select the DHCP option instead of Static. The message "DHCP request sent" appears briefly, and each PC then displays its newly assigned IP address, subnet mask, and DNS server (192.168.10.1).

▪ Verify the DHCP configuration by opening the Command Prompt of any PC and entering the command ipconfig /all. The output shows the IP address received from the pool (starting from 192.168.10.2), the subnet mask, and the DNS server address. Each PC receives a different address.

▪ Verify connectivity with the ping command. From a PC, enter ping 192.168.10.1. Four successful replies with 0% loss confirm that the PC can reach the server. Also ping between two PCs, for example from STU1 to FAC1, using the DHCP-assigned address of FAC1.

▪ Test DNS resolution from the Command Prompt using nslookup lab.college.edu and ping lab.college.edu. Both resolve the name to 192.168.10.1, which confirms that DNS is working correctly.

▪ Test the FTP upload by the faculty user. On FAC1, open Desktop → Text Editor, type some text, and save the file as lab1_manual.txt. Then open the Command Prompt and enter the following.

```
C:\> ftp lab.college.edu
Username: faculty1
Password: Fac@123
ftp> put lab1_manual.txt
```

ftp> dir

ftp> quit

▪ A “Transfer complete” message confirms the upload, and the file lab1_manual.txt now appears in the server’s file list under Services → FTP.

▪ Test the FTP download by the student user. On STU1, open the Command Prompt and enter the following.

C:\> ftp lab.college.edu

Username: student1

Password: Stu@123

ftp> dir

ftp> get lab1_manual.txt

ftp> quit

▪ The file is listed and downloaded successfully, which confirms that students can access the resources uploaded by the faculty.

▪ Perform the permission test (negative test). While logged in as student1, enter put anyfile.txt. The server denies the upload because this account has only Read and List permissions, which proves that the access control works.

▪ Also try logging in with a wrong password. The login fails, which confirms that only authorized users can access the FTP server.

▪ The network can also be tested in Simulation Mode by filtering DHCP, DNS, and FTP to observe the packets passing through the switch.

▪ Screenshots of the topology, the server’s static IP configuration, the DHCP pool, the DNS record, the FTP accounts, the ipconfig /all output, the ping results, the nslookup result, and the FTP upload, download, and denied-upload tests are captured as evidence of the completed setup.

▪ Finally, the completed Packet Tracer project is saved as a .pkt file, along with the required documentation and screenshots for submission.

Conclusion

The college computer laboratory network was successfully designed and configured in Cisco Packet Tracer, demonstrating how DHCP, DNS, and FTP can be combined on a single central server. The experiment involved configuring the server with a static IP address, creating a DHCP pool for automatic addressing, adding a DNS record for the domain name lab.college.edu, and creating FTP accounts with different permissions for faculty and students. Connectivity was verified using the ping command, name resolution using nslookup, and file sharing through FTP upload and download tests, including a negative test in which the student account was denied permission to upload. This experiment reinforced the practical understanding of DHCP, DNS, FTP, access control, and centralized network services in a laboratory network.